Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/13/2012
PI Name:	Redpath, John Leslie Ph.D.		
Project Title:	High energy proton dose-rate and mixed field effects on neoplastic transformation in vitro		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jlredpat@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-7395
Organization Name:	University of California Irvine		
PI Address 1:	Radiation Oncology		
PI Address 2:	School of Medicine, Med Sci I, B148		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 12/31/2011 (from 12/31/2010) per C. Guidry/JSC (Ed., 2/3/11)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Elmore, Eugene (University of California Irvine)		
Grant/Contract No.:	NNJ06HD90G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The central objectives are 1) to determine the the factor by which the effectiveness of high energy protons is reduced when they are delivered at dose-rates that correspond to solar particle events, and 2) to determine the effectiveness of mixed fields of protons and heavy ions compared to either alone. The cancer-relevant endpoint of neoplastic transformation in vitro will be used.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The research carried out under this award is of relevance to the estimation of cancer risks involved in space exploration. The experimental focus was on cancer risk following exposures to high energy protons delivered at low dose rate, as well as to mixed fields of protons and iron ions. The approach was to study the radiation-induced conversion of human-derived cells in vitro from a non-cancerous to a cancerous phenotype. The aims addressed NASA's research emphases with respect to reductions in cancer risk projection uncertainties.
Task Progress:	Background: This research project had two major goals. The first goal dealt with dose-rate effects for high energy proton irradiation and these experiments were carried out at the Loma Linda University High Energy Proton Facility. In this regard we hypothesized that at the peak dose-rate of a major solar particle event (SPE) of around 200 mGy/h (3.3 mGy/min) the threshold dose for neoplastic transformation by high energy protons will be >100 mGy. We further hypothesized that the slope of the dose-response curve at doses above the threshold would be less that previously observed with high dose-rate protons (1). The second goal dealt with mixed beam (proton plus iron) effects and these experiments were carried out at the NSRL at Brookhaven National Laboratory. In addition to mixed beam experiments we also carried out dose response curves for each of particle alone. For the sequence protons followed by heavy ions we hypothesized that an adaptive response would be induced by the protons which could potentially ameliorate the effect of the iron ions, particularly at iron doses >100 mGy. This adaptive response would, however, be dependent on the time interval between treatments. For iron ions followed by protons we hypothesized that this adaptation would be largely absent and an essentially additive effect would be seen at proton doses >100mGy. Results High energy proton dose-rate effects. We ran several experiments at Loma Linda designed to determine the dose and dose-rate effectiveness effectiveness factor (DDREF) for high energy (232 MeV). The combined data indicated no difference between the two dose-rates (20 cGy/h and 50 cGy/min) in induction of transformation following doses in the range 5 to 100 cGy, indicating a DDREF of 1 over this dose range for this endpoint. If this finding were to hold for other endpoints related to cancer induction it could mean that high dose-rate data for doses up to 100 cGy may be enough to estimate the relative risk of carcinogenesis at the highest dose-rates associated with solar flares. This work was published in the June 2011 issue of the International Journal of Radiation Biology (2). While at NSRL we also ran assays to complete a dose-response curve for 1 GeV/n protons for comparison with the Loma Linda data. The NSRL findings confirm our earlier report of possible suppression of neoplastic transformation at doses <10 cGy using 232 MeV/n protons from Loma Linda. They also indicate that at a dose of 100 cGy the 1 Gev/n protons may be more biologically effective than the 232 MeV/n protons. Dose-response for 1GeV/n iron ions. Early studies in the award period indicated a threshold-type dose-response curve for neoplastic transformation induced by 1 Gev/n iron ions with a dose threshold of 5 to 10 cGy (3). This was an unexpected finding. We then further examined this response at doses down to as low as 0.01 cGy and now have strong evidence for suppression of transformation at doses <1 cGy (4). We have postulated that the mechanistic explanation for this observation lies in the fact that at doses <100 mGy the majority of cells are not traversed by an iron ion track, and the low-LET delta ray penumbra extends several cell diameters imparting doses of up to 10 mGy, it is feasible that the low-LET delta ray component may be suppressing the inducing effect of the iron through an adaptive response. Mixed beam studies. The mixed beam experiments were to test the effect of 10 cGy of 1 Gev/n protons combined with 100 cGy of 1 GeV/n iron and 10 cGy of 1 GeV/n iron followed by 100 cGy 1 GeV/n protons. In each case the second dose was delivered either immediately (<15 min), or after a 24 h delay, following the first dose. For low dose protons followed by high dose iron there was no evidence of an adaptive response with the results being statistically consistent with additivity. Surprisingly, for low dose iron followed by high dose protons there was an apparent adaptive response. In an attempt to explain the data, we used the same protocol to test for an adaptive response using the 232 MeV/n proton beam at Loma Linda (i.e. low dose followed by high dose protons) and similarly using the Cs-137 gamma source at U.C. Irvine. Again, we did not see any evidence for an adaptive response. Since an adaptive response should have been seen with these qualities of radiation this led us to question the experimental protocol and we reduced the interval between adapting and challenge dose from 24h to 8h and tested this using Cs-137 gamma radiation. This resulted in a positive adaptive response being identified and indicates that the interval used in the mixed beam was too long for such a response to be seen. Nonetheless, the findings are useful and are indicative that temporal separation of traversal of a cell by one particle type and then another has to be quite short for any adaptive response to be seen and this scenario may be quite unlikely in the space environment. These experiments and assays took up most of 2010 to complete and this study was written up earlier in 2011 year and subsequently published in Radiation Research (4). References (note: #’s 2, 3, and 4 are a direct result of this award). 1. E. Elmore, X.Y. Lao, M. Ko, S. Rightnar, G. Nelson and J.L. Redpath, Neoplastic transformation in vitro induced by low doses of 232 MeV protons. Int. J. Radiat. Biol. 81: 291-298, 2005. 2. E. Elmore, X-Y Lao, R. Kapadia and J.L. Redpath, Effect of dose-rate on induction of neoplastic transformation in vitro by low doses of 232 MeV protons. Int. J. Radiat. Biol. 87: 622-627, 2011. 3. E. Elmore, X-Y Lao, R. Kapadia and J.L. Redpath, Threshold-type dose response for induction of neoplastic transformation by 1 GeV/n iron ions. Radiation Research 171:764-770, 2009. 4. E. Elmore, X-Y Lao, R. Kapadia, M. Swete and J.L. Redpath, Neoplastic transformation by mixed beams of high energy iron ions and protons. Radiation Research 176: 291-302, 2011.
Bibliography Type:	Description: (Last Updated: 01/13/2012)

Articles in Peer-reviewed Journals	Elmore E, Lao XY, Kapadia R, Swete M, Redpath JL. "Neoplastic transformation in vitro by mixed beams of high-energy iron ions and protons." Radiat Res. 2011 Sep;176(3):291-302. Epub 2011 Jul 6. PMID: 21732791 , Sep-2011